

PIYUSH SHARMA

Enterprise Automation Lead | AI-Driven Systems Driving Transition from Enterprise Automation to AI-Enabled Systems

📞 +91-8904767033 📧 inboxofpiyush@yahoo.com 🔗 <https://www.linkedin.com/in/piyush-sharma-4b6137216>

🔗 <https://piyush-sharma-portfolio.vercel.app> | <https://github.com/inboxofpiyush01> 📍 Delhi, India

⭐ Open to relocation : UAE, UK, Germany, Netherlands, Ireland, Singapore, Australia



PROFILE SUMMARY

Results-driven Enterprise Automation Lead with 9+ years of experience in solution design, with expertise in VBA and Python scripting, delivering scalable solutions across Banking, Insurance, Healthcare, and Mortgage domains. At Mphasis, led development of 250+ automation tools, streamlining operations and generating \$5M+ in business benefits for clients including JPMC, Charles Schwab, and Zurich. At KPMG, served as Assistant Manager – Technical, developing VBA and Python-driven automation, maintaining SQL databases, and enhancing audit and advisory processes. Recognised with 16+ performance awards including Rising Star, Senior of the Quarter, and Summit Awards across KPMG and Mphasis. Experienced in workflow automation, API integration, and mentoring project teams. Currently upskilling in Artificial Intelligence and Generative AI to build intelligent, data-driven systems that integrate automation with predictive analytics.

EXPERIENCE

Independent AI Engineering & Automation Consulting

Generative AI & Intelligent Automation Projects

📅 01/2025 - Present 📍 Delhi, India

Structured transition programme from enterprise automation into AI engineering, with projects deployed on GitHub.

- Completed online certifications in AI and Machine Learning to strengthen foundational knowledge.
- Pursuing a structured Generative AI Master Program alongside hands-on project development to transition from enterprise automation to AI-enabled systems.
- Gaining practical exposure to Agentic AI workflows and orchestration frameworks
- Understanding MLOps concepts including model lifecycle, deployment, and monitoring
- Gained hands-on experience with LangGraph, LangChain, CrewAI, HuggingFace, Streamlit, FAISS, Scikit-learn, MLflow and Docker
- Exploring supervised learning techniques (classification, regression) and NLP pipelines for text analysis and sentiment classification.
- Actively building GitHub portfolio at github.com/inboxofpiyush01

Assistant Manager - Technical

KPMG

📅 06/2023 - 12/2024 📍 Bangalore, India

KPMG is one of the Big Four global professional services firms delivering Audit, Tax, and Advisory services. KPMG India focuses on technology-driven solutions to enhance business assurance and operational efficiency.

- Developed and deployed automation bots using Python for Tech Assurance processes, reducing manual efforts by 30–40% and improving process accuracy.
- Created VBA-based tools to automate and integrate third-party application APIs, streamlining audit and assurance workflows and saving significant project time.
- Designed and maintained SQL databases for multiple assurance projects, ensuring data consistency, reliability, and optimized query performance.
- Worked closely with audit and assurance teams to translate business requirements into automation solutions, enhancing efficiency and reporting accuracy.
- Provided technical guidance to cross-functional teams on automation tools and contributed to process documentation and best practices.
- Tools & Technologies:** Python, LLM, VBA, SQL Server, APIs, MS Excel

KEY ACHIEVEMENTS



Financial Achievement

Achieved benefits of more than **\$5M** through automation solutions



Performance Awards : Total: 16+

KPMG Recognition (2023–2024, Total: 6 Awards): Received **4 Encore/Kudos Spot Awards, Senior of the Quarter, and Rising Star Award** for outstanding performance, leadership, and technical contributions.

Next Ops RNR Award (Mphasis, 2022,

Total: 1 award): Recognized for exceptional client support and contribution to business outcomes.

Demanding Excellence Awards (Mphasis,

2020–2021, Total: 2 awards): For consistently demonstrating Mphasis values through high-quality delivery and teamwork.

Agility Is My Mantra Awards (Mphasis,

2017–2020, Total: 4 awards): For outcome-focused execution, effective problem-solving, and ownership of decisions.

The Buck Stops with Me Awards

(Mphasis, 2016–2019, Total: 4 awards):

For customer-centric solutions, accountability, and innovative approaches.

Summit Awards (Mphasis, 2016, 2017,

Total: 2 awards): For overall excellence and high-performance contribution across projects.



Academic Achievements

B.Tech. Academic Excellence – Secured **topper position in all four years** of the program.

Six Sigma Black Belt Certification –

Achieved **Benchmark Seal of Excellence** for highest exam score.

Embedded Systems & Robotics –

Successfully qualified the examination conducted by **International Association for Distance Learning**.

ENTERPRISE AI & AUTOMATION EXPERTISE

- Intelligent Automation
- RAG Systems
- Multi-Agent AI
- Workflow Orchestration
- API Integrations
- Enterprise Process Automation
- AI-assisted Documentation
- Audit & Compliance Automation
- LLM Prompt Engineering & Context Management
- AI-Powered Knowledge Retrieval Systems
- Conversational AI & Enterprise Chatbots
- Automation Solution Architecture & Technical Design

EXPERIENCE

Senior Software Engineer/Module Lead

Mphasis India Pvt. Ltd

11/2015 - 06/2023 Pune, Bangalore India

Mphasis is a global IT solutions provider serving Fortune 500 clients across Banking, Insurance, Healthcare, and Mortgage domains, leveraging automation, AI, and analytics for digital transformation.

- Led a team of 21 developers across multiple projects, ensuring effective collaboration, reduced resource wastage, and improved delivery reliability.
- Collaborated with client Business & IT teams to gather, refine, and translate requirements into scalable technical solutions.
- Designed solution architectures and acted as Lead Architect, owning end-to-end design and implementation.
- Performed technical QA reviews, code documentation checks, and project quality assurance activities.
- Developed test cases and troubleshooting procedures for unit and integration testing; enhanced system stability and data visibility.
- Conducted process improvement sessions (Day-in-Life) across business accounts to identify automation opportunities.
- Mentored project teams on VBA, Python, and data handling best practices.
- Managed client coordination, requirement analysis, and project governance reviews.
- Led automation initiatives integrating applications (Web, MS Office, Outlook, Mainframe) using VBA and workflow tools.
- Defined and implemented end-to-end RPA and automation strategies to improve process efficiency.
- Technologies: Python, VBA, SQL Server, APIs, MS Office, Workflow Automation, RPA Tools

KEY PROJECTS

Multi-Agent Enterprise AI Assistant

2026 Delhi, India

https://github.com/inboxofpiyush01/Universal_AI_Chat_Bot

Multi-Agent AI Chatbot Platform (SaaS-Oriented)

- Plug into any website — the bot learns from whatever is on that website automatically.
- Implemented contextual response handling using structured prompts.
- Developed architecture supporting multi-agent interaction using CrewAI.
- Tools: Python · LLM · RAG · CrewAI · BeautifulSoup · Visual Studio

Enterprise RAG Knowledge Assistant

2026 Delhi, India

https://github.com/inboxofpiyush01/doc-qa-assistant

Upload any PDF or text document and ask questions in natural language. Powered by Retrieval-Augmented Generation (RAG) using free open-source models.

- Developed a RAG-based document Q&A system using LangChain & FAISS, enabling faster document insights and reduced manual effort
- Implemented semantic search with MiniLM embeddings for accurate and efficient information retrieval
- Integrated FLAN-T5 LLM for context-aware responses with source attribution, enhancing answer reliability.
- Tools: Python · LangChain · RAG · Streamlit · HuggingFace · Visual Studio

Geo-Intelligence Sales Analytics System

2025 Delhi, India

https://github.com/inboxofpiyush01/ClientAnalysis

Analysed a dataset of hotels and restaurants to find the best potential buyers for a floor cleaner product using Python — covering data cleaning, distance calculation, and visual reporting.

- Cleaned and prepared a messy real-world business dataset by removing unnecessary columns, filtering out closed and unverified businesses, and fixing missing values
- Calculated how far each business is from the company's office using GPS coordinates, and shortlisted only the nearby ones as potential sales targets
- Created charts to visually understand which types of businesses are most common, how ratings are distributed, and which customers have the best reviews and ratings
- Identified the top 10 highest-rated, most-reviewed nearby businesses and exported them to an Excel file ready for the sales team to use
- Tools: Python · Pandas · NumPy · Matplotlib · Seaborn · Jupyter Notebook

CERTIFICATION/TRAINING

Agentic AI, Artificial Intelligence & MLOps Certification Course

Edureka, online

Python 3.X, Data Science, AI/ML

Simplilearn in collaboration with IBM, online

Complete Data Science, ML, DL, NLP Bootcamp

Udemy, online

Machine Learning

Coding Bytes, Delhi, India

Six Sigma Black Belt Certification

Benchmark Six Sigma, Bangalore, India

Excellence in Continuous Improvement Leadership Certification

Benchmark Six Sigma, Bangalore, India

SKILLS

Languages & Scripting:

Python, VBA, SQL

AI / ML / GenAI:

LLMs, LangGraph, LangChain, CrewAI, RAG, LLM Prompt Engineering, Generative AI, Agentic AI, NLP, Machine Learning, Deep Learning, Data Science

ML Libraries:

Scikit-learn, Pandas, NumPy, TensorFlow

MLOps:

MLflow, Docker, Git, GitHub, Streamlit

AWS:

EC2, S3, IAM

Automation & RPA:

RPA Tools, MS Office Automation, API Integration, Workflow Automation, MS Excel

Databases:

SQL Server, MS Access, MongoDB

EDUCATION

PGDM in Business Management

Amity School of Distance Learning

01/2012 - 05/2015 Delhi, India

B.Tech in Electronics & Communication

Delhi Institute of Technology and Management

08/2008 - 05/2012 Delhi, India

LANGUAGES

English

Proficient



Hindi

Native



KEY PROJECTS

QA Dummy Data Generator

📅 2024 📍 KPMG, Bangalore, India

Built an internal QA productivity tool at KPMG that auto-generates realistic test datasets with multiple scenarios — solving the compliance problem of QA teams not being able to use real client data for testing.

- Tool automatically detects the input file structure (type, columns, rows) and asks QA to simply specify requirements like number of records, columns to include, data points to randomise, and header preferences, etc.
- Auto-generates a structured prompt from these inputs and feeds it into KPMG's audit AI tool, which then produces ready-to-use dummy datasets — saving hours of manual data creation per test cycle
- Developed first in VBA for quick rollout, later rebuilt in Python for enhanced flexibility and wider adoption across teams
- **Tools:** VBA, Python, Prompt Engineering, MS Excel, MS Office Automation

BRD Generator / Translator

📅 2024 📍 KPMG, Bangalore, India

Tool that reads code or an existing BRD and converts it into simple non-technical language — helping Business Analysts understand project requirements without any technical background.

- Reads code or BRD documents and auto-generates a structured prompt, fed into KPMG's internal audit AI to produce readable business documentation
- Eliminated manual BRD writing effort — BA teams could understand requirements without needing technical knowledge
- Built in VBA first for quick deployment, then rebuilt in Python for better scalability.
- **Tools:** VBA · Python · Prompt Engineering · MS Office Automation

KCAT Tool

📅 2023 📍 KPMG, Bangalore, India

Automated end-to-end audit confirmation process, including authentication and account management via APIs.

- Reduced processing time and ensured accurate tracking of all confirmations for auditors.
- **Tools:** VBA · Web Application · API · SQL · MS Excel · MS Outlook

PowerBI Elements Extractor

📅 2023 📍 KPMG, Bangalore, India

Automated extraction and validation of tables, queries, and formulas from PowerBI dashboards.

Converted backend JSON data into tabular format for easy QA validation.

- Reduced QA effort significantly and improved accuracy of dashboard QC checks.
- **Tools:** VBA · PowerBI · MS Excel

PRE-OPS Automation Tool

📅 2020 📍 Mphasis, Pune/Bangalore, India

Reduced mortgage QC processing time significantly by automating end-to-end data fetch and validation across multiple applications

- Automated the End-to-End Mortgage QC process
- Developed a tool that fetches and processes data from various applications
- **Tools:** VBA · Web Applications · Mainframes · MS Excel · MS Outlook

Gifting Tool

📅 2019 📍 Mphasis, Pune/Bangalore, India

Eliminated manual share transfer processing, reducing errors and saving team hours per week

- Created a VBA Solution to automate share transfers for clients
- **Tools:** VBA · Web Applications · Mainframes · MS Excel · MS Outlook

ANNEXURE

Kindly refer to annexure for some other high impact projects:

- ❖ LIBOR Validation Tool, Client: JPMC LIBOR, Bangalore
- ❖ Auto Faxing Tool, Client: Charles Schwab, Pune & Bangalore
- ❖ UGL Recon Tool, Client: Charles Schwab, Pune
- ❖ Notes- Standardization Tool, Client: Charles Schwab, Pune & Bangalore
- ❖ Mistake Proofing Tool, Client: Virgin Media-UPC, Pune & Bangalore
- ❖ KDAT Tool, Client: Charles Schwab, Pune & Bangalore
- ❖ Lean Assessment Tool, Client: Charles Schwab, Pune & Bangalore
- ❖ Workflow Management Tool, Client: JPMC, Pune & Mumbai
- ❖ MIS & Notes Standardization Tool, Offshore
- ❖ EOS Report – Pune, Offshore
- ❖ Self QC Tool, Offshore
- ❖ Auto Fetching of Account, Job ID from Application to Client Macro, Offshore
- ❖ Auto Fetching of Journal Effective Date in Client Macro, Offshore
- ❖ Account Closure Automation, Offshore
- ❖ TraQer Updation Tool, Offshore
- ❖ ACATS Mistake Proofing Tool, Offshore
- ❖ Signature Verification Tool, Offshore
- ❖ Wire Transaction - Mistake Proofing Tool, Offshore
- ❖ Fund Changes Recon Tool, Onshore
- ❖ TV Address Recon Tool, Onshore
- ❖ ANF Dividend Tool, Onshore
- ❖ Directive Recon - Daily Aged Cash Report, Onshore
- ❖ Pending Batch Recon Tool, Onshore
- ❖ ETF - Distributions Tool, Onshore
- ❖ Auto Letter Creation Tool, Onshore
- ❖ Loan Out of Balance Tool, Onshore
- ❖ Auto Time Zone, Offshore
- ❖ MBMS (Mailbox Management System), Client: Star India, Pune & Mumbai